Saurav Sana

Engineering Lead & Solution Architect | Publicis Sapient | Ex-PwCI, Ex-Samsung

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Engineering Lead and Solution Architect with over **20 years** of experience designing, building and operating data-intensive, multi-tenant platforms on Azure and AWS, most recently a global regulatory reporting platform processing 10+ million messages a day for 100+ clients. Pairs hands-on engineering depth (firmware in C through cloud-native microservices and lakehouse platforms) with an evidence-driven approach to architecture: SLA/SLO design, queueing and probability models, formal verification (TLA+) and FinOps.

- **Solution architecture and modernization:** decoupling monoliths into microservices with Domain-Driven Design and phased roadmaps, aligned with the well-architected framework and with minimal end-user impact.
- **Cloud, data and security:** cloud adoption, security, integration, data engineering (Apache Iceberg lakehouse, Spark), automation and full-stack enterprise solution development.
- **Engineering leadership:** led teams of up to 25 engineers to deliver resilient, high-performing, scalable, data-intensive, multi-tenant compliance applications, with design reviews, quality gates and release-readiness practices.

Notable Achievements :

Engineering Lead with strong domain expertise in financial regulatory reporting, known for transforming vision into production-ready platforms. Proven at navigating constantly evolving global regulatory landscapes to deliver secure, scalable, multi-tenant cloud solutions that ensure client compliance, reduce risk and cost, and provide unified OTC trade reporting across asset classes and jurisdictions.

- Converts **regulatory vision and business strategy into end-to-end products**, from concept through scalable, compliant production systems.
- Expert at **operating within rapidly changing regulatory environments**, ensuring clients remain compliant as the platform grew from **2 to 10+ regimes in 3 years**.
- Designed and delivered a scalable, reliable multi-tenant platform that processes over **10 million messages daily**, delivering to **10+ regulators within five minutes**, with a **near real-time analytics dashboard** and efficient search and analytics over the **last ten years** of submissions from **100TB** of data at minimal cost.
- Raised trade-processing SLA from **~80% to 99.9% in 18 months** while cutting infrastructure cost **~50% ($600K to $300K per month)**: reliability and cost improved together.
- **Independent author** of two engineering series: **The Shape of Failure** (probability and failure modes in distributed and AI systems) and **Beyond the MPP Default** (Iceberg data-gateway architecture, formally verified with TLA+).

Educational Background

Indian Institute of Technology, Kharagpur
B.Tech | 2006
- Major in Electrical Engineering
- Minor in Electronics & Telecommunications

R.K. Mission Residential College, Narendrapur
H.S | 2001

Technical Certifications

- The Open Group | TOGAF 9.2
- Microsoft
  - Azure Solution Architect Expert
  - Azure DevOps Engineer Expert
  - Azure Developer
- AWS
  - Solution Architect Associate
  - Solution Architect Professional
  - Security Specialty
- Stanford Online
  - Machine Learning

Professional Career

Publicis Sapient | Financial Service North America
- **Senior Manager Technology** | Since *June 2022*
  Engineering Lead | DTCC/DC, Financial Regulatory Reporting

Independent | Field Notes (technical writing)
- saurav-sana.github.io/blog

PwC India | MS Enterprise Application
- **Associate Director** | *April 2019 to May 2022*
  Solution Architect, Delivery Lead, Emerging Technologies
- **Principal Consultant** | *April 2016 to March 2019*
  Solution Architect, Development Lead, Drone Solution
- **Senior Consultant** | *April 2010 to March 2016*
  Development Lead, Full Stack Developer
- **Consultant** | February 2009 to March 2010
  Software Developer

Samsung India | Memory Solution Division
- **Senior Software Engineer** | July 2006 to January 2009
  Firmware & Embedded system developer

Key Skills, Tools & Programming Languages

Enterprise Solution Architecture | Engineering Leadership | Emerging Technologies | Data Engineering | Azure | AWS | Kubernetes | .NET | Microservices | APIs | Data Lakehouse | Spark | Apache Iceberg | Dremio | Observability | SRE and SLO Design | FinOps | Formal Verification (TLA+) | OAuth, SAML | Python | JavaScript | C/C++ | DevOps

Engineering lead - Roles and Responsibilities

- **End-to-end ownership of a global, multi-tenant regulatory reporting platform**: architecture, design, build, scale and production reliability, across 100+ clients and 10+ regulatory regimes.
- Designed and evolved a **cloud-native, microservices architecture on Azure**, ensuring high availability, fault tolerance, tenant isolation, security and regulatory compliance.
- Drove **platform performance and SLA excellence**, meeting strict regulatory requirements for timeliness, completeness and correctness at high scale, and addressed technical debt to improve **stability, reliability and cost optimization**.
- Led **data platform modernization**: legacy data migrations, unified data models and analytics-ready architectures, enabling near real-time ingestion and analytics on a **Data Lakehouse**.
- Established **engineering and reliability practices:** design reviews, code quality, observability and release readiness.
- Led and mentored cross-functional engineering teams across backend, cloud, data and infrastructure, enabling faster delivery while keeping the **platform resilient and scalable**.
- Owned **cost optimization (FinOps)**: reduced infrastructure spend and manual support activity through architecture, data-lifecycle and automation initiatives.
- Explore opportunities to incorporate AI features into the product for better user experience and value, including a **proactive and reactive autonomous operational agent**.

Experience

Publicis Sapient, Remote Working | Senior Manager Technology – Since June 2022

DTCC/DC ReportHub for Financial Regulatory Reporting -

Report Hub addresses the growing complexity of global trade reporting by providing clients with a single, centralized platform to manage regulatory compliance across jurisdictions. It enables financial institutions to ingest, validate, enrich, reconcile and submit high-volume trade data accurately and on time, replacing fragmented systems and manual processes that increase operational risk. By enforcing standardized reporting, strong data quality controls and end-to-end auditability, the platform significantly reduces regulatory breaches and compliance effort. The scalable, secure, cloud-native design lets clients support new regulations and growth while keeping near real-time visibility into reporting status and data health.

- Achieved **~50% reduction in platform infrastructure cost**, lowering monthly spend from **$600K to $300K over 3 years** through architecture optimization, cloud-efficiency improvements and data-lifecycle tuning.
- Improved trade-processing SLA from **~80% to 99.9% within 18 months**, consistently meeting strict regulatory requirements for **timeliness, completeness and correctness** under high-volume trades.
- Designed and scaled a **high-throughput platform** processing **10+ million messages per day at up to ~20K events/second**, with zero production failures.
- Engineered **30% burst headroom for volatile market conditions** on the same infrastructure footprint.
- Built a secure, **distributed, fault-tolerant multi-tenant SaaS platform serving 100+ global clients**, with strong tenant isolation, platform security and data-confidentiality controls across regions.
- Expanded platform capabilities from **2 to 10+ global regulatory regimes** in 3 years, enabling faster regulatory onboarding and sustained business growth.
- Led **large-scale data migration** from multiple legacy platforms into a unified architecture, improving data reliability, auditability and retrieval efficiency.
- Delivered a modern **Data Lakehouse in 9 months using Apache Iceberg**, .NET ingestion services and enterprise PySpark, resulting in:
  - **30% reduction in storage costs**
  - **Near real-time analytics** and search dashboard with **P95 latency under 10** seconds
- Built a **centralized observability and reliability platform** using **Grafana and Prometheus**, enabling proactive issue detection, root-cause analysis and SLA protection.
- Reduced support effort and operational overhead by **30% over 2 years** through platform optimization, automation, improved observability and stability enhancements.

Architecture and engineering leadership highlights –

- **Regulatory change to roadmap:** translated evolving regulations and business strategy into platform capabilities, so new regimes were onboarded on the same architecture rather than rebuilt.
- **SLA-driven engineering:** timeliness, completeness and correctness run as measurable service levels, protected by observability, proactive alerting and root-cause analysis.
- **Technical debt as a reliability and cost lever:** addressed technical debt to improve stability and reliability while optimizing run cost and manual support effort.

Technology: Azure | Microservices | .NET | PySpark | Apache Iceberg | Data Lakehouse | Grafana | Prometheus

Independent Technical Writing

Field Notes | saurav-sana.github.io/blog

Independent engineering blog on distributed systems, queueing physics and enterprise software economics, applying probability, queueing theory and formal methods to real architecture decisions.

- **The Shape of Failure** | 8-part series plus AI/LLM appendix (Oct 2026): a field guide to why distributed systems break in unexpected ways. It maps 20+ probability distributions to the engineering question each answers (reliability, cascading failures, extreme latency, data skew, retries and queues), with modelling, estimation and engineering controls, extended to AI/RAG systems, LLM infrastructure and FinOps.
- **Beyond the MPP Default: Designing a Data Gateway for the 95% Reality** | 5-part series (Sep to Oct 2026): an architecture blueprint that turns immutable Iceberg lakehouses into interactive, zero-copy query endpoints for web APIs and AI agents (REST, gRPC, MCP), bypassing MPP engines for small analytical queries. Covers HA, SRE and FinOps modelling, **TLA+/PlusCal formal verification of the design (model-checked with TLC)**, and production architecture, multi-agent platforms and build-vs-buy trade-offs.

PwC India, Kolkata | Career Growth

Consultant (2009) | *Senior Consultant* (2010) | *Principal Consultant* (2016) | *Associate Director* (2019 – *May* 2022)

Application Modernization and Cloud Architecture –

Led modernization of multiple Risk, Independence and Tax platforms, aligning technology with business strategy while managing **20–25 engineers across India**. Designed and delivered enterprise-scale, resilient and scalable cloud solutions with strong focus on security, compliance and automation.

- Drove **monolith-to-microservices transformation using Domain-Driven Design** with phased roadmaps and minimal end-user impact. Re-platformed legacy integration and UI systems (BizTalk, SharePoint) to Azure VM clusters, improving performance, security and observability.
- Established **engineering excellence practices**: design and code reviews, CI/CD with quality gates, DR/BCP planning with regular drills and proactive configuration management. Built self-service rule platforms, event-driven messaging and modern UX with strong authentication and authorization.
- Delivered **hybrid-cloud data aggregation** processing daily global broker feeds to keep portfolios compliant across regional regulations.

Drone Powered Solutions -

Ideation of a drone solution ensuring regulation-compliant drone operation and helping clients in TA activity with digital transformation for operational efficiency and transparency, using UAV, ML, Data Science, IoT and cloud solutions.

- Presented a prototyped **precision-farming solution** at the Essar Data Science Congress 2018.
- TA activities for **blood-sample delivery by drone** to remote locations for a state government of India.
- Prototype solutions for **surveillance, volumetric analysis and asset tracking** for open-cast mining operations.

Full Stack Developer –

Extensive full-stack development across the Microsoft stack, MEAN and MERN, spanning front-end, backend, data engineering, system integration, rules engines, databases and reporting platforms, with practical exposure to Linux system administration and platform operations.

Served as **Lead Developer on secondment to PwC US**, driving requirements, technical design, integration, infrastructure planning and delivery using .NET, SQL Server and BizTalk. Built event-driven middleware, tax and document-repository systems with integrations, automation, monitoring, reconciliation and admin portals.

Delivered diverse solutions across industries and geographies, including:

- Hybrid and enterprise platforms for regulatory, tax and compliance use cases; CMS with secure payment-gateway integration (Drupal + PayPal) for CSR initiatives
- Low-level, high-performance systems such as Call Detail Record decoders in C, integrated with Oracle Data Warehouses
- Configurable rules engines and web services for tax calculations and external client integrations
- Desktop productivity tools integrated with Documentum APIs for large-scale document-management workflows

Senior Software Engineer | Samsung India Software Operations, Bangalore | Jul 2006 – Jan 2009

Worked as a firmware and device-driver programmer on the Linux platform, building solutions to optimise boot time, application run time and file-system access for Flash memory devices.

- In-depth knowledge of Linux, device drivers and bootloaders. **Optimised boot time on Samsung's flagship OneNAND flash drive**, ported and tuned the MTP protocol in Samsung's Linux OS, and wrote ARM assembly in the bootloader.
- Fluent in C and C++: low-level coding with efficient algorithms and data structures for performant drivers and systems.